TWO-ENDS GEOMETRY AND MULTILINEAR METHODS IN FOURIER RESTRICTION

XIANGYU WANG

Abstract. We study Fourier restriction estimates in dimensions $n \geq 5$. The paper has two parts. First, we refine the decoupling–incidence method of Wang-Wu. Second, we develop a novel and concise framework for proving restriction estimates, which yields stronger results than those obtained by polynomial partitioning methods in higher dimensions.

1. Introduction

1.1. **The problem and the two parts of the paper.** Let B^{n-1} be the unit ball in $\mathbb{R}^{n-1}$, and define

$$Ef(x', x_n) = \int_{B^{n-1}} e^{i(x'\cdot\xi + x_n|\xi|^2)} f(\xi)\, d\xi. \tag{1.1}$$

The diagonal restriction problem asks for the range of p for which

$$\|Ef\|_{L^p(\mathbb{R}^n)} \leq C_{n,p}\|f\|_{L^p(B^{n-1})}. \tag{1.2}$$

The conjectured range is $p > 2n/(n-1)$. Throughout this paper, n denotes the *ambient* dimension, the input and output exponents are the same, and $n \geq 5$. We do not address the cone or variable-coefficient operators.

The paper has two complementary parts. The first retains the architecture of Wang–Wu [WW]: a two-ends reduction separates a scale-inductive contribution from a geometrically dispersed contribution; local L^2 orthogonality and refined decoupling then convert tube multiplicity bounds into restriction estimates. We reconsider the available incidence bounds before eliminating their density parameters. This yields a small high-dimensional improvement, including a strict improvement over the explicit benchmark in dimensions $18 \leq n \leq 21$, $n = 23$, and $n \geq 25$. The leading high-dimensional constant is unchanged.

Theorem 1.1 (Refinement)**.** *Let $18 \leq n \leq 21$, $n = 23$, and $n \geq 25$. Then the estimate* (1.2) *holds for*

$$p > 2 + \frac{4(7n+15)}{11n^2+10n-3}$$

Compared with the Wang–Wu exponent

$$P_{\mathrm{WW}}(n) = 2 + \frac{4(7n+29)}{(n-1)(11n+43)}, \tag{1.3}$$

this refinement yields a further improvement of

$$\frac{72(5n+31)}{(n-1)(11n+43)(11n^2+10n-3)} > 0. \tag{1.4}$$

The second part develops a novel framework for proving Fourier restriction estimates. Zahl's direction-separated multilinear Kakeya estimate retains geometric information that is lost when passing to its linear Kakeya maximal counterpart. Rather than linearizing this estimate, we incorporate it directly into a broad–narrow argument. The resulting approach is concise and yields stronger restriction exponents than those obtained by existing polynomial partitioning methods in higher dimensions.

This construction is motivated by the incidence organization behind planebrush arguments. The four-dimensional developments of Katz–Zahl [KZ], Borges–Chan–Chen–Liu–Xi–Zhan [BCCLXZ], and Moreira [Moreira] illustrate the value of retaining intermediate multiplicities instead of replacing them immediately by total tube counts. A corresponding unrestricted planebrush estimate is not supplied here in higher dimensions. Our present substitute is to retain a genuinely transverse multilinear configuration, for which the geometric input is available. Thus the new framework is *planebrush-inspired*, not a claim that the four-dimensional planebrush theorem extends verbatim to every n.

Fix

$$2 \le k \le \left\lfloor \frac{n+2}{2} \right\rfloor, \qquad d = d_{n,k}, \qquad D = (4n-5)(d-1) + (n-1)(n-3). \tag{1.5}$$

Define

$$p_b(n,k) = 2 + \frac{4(2d+n-5)}{D}, \tag{1.6}$$

$$p_N(n,k) = 2 + \frac{4}{2n-k}, \tag{1.7}$$

$$P_{\mathrm{II}}(n) = \min_{2 \le k \le \lfloor (n+2)/2 \rfloor} \max\{p_b(n,k), p_N(n,k)\}. \tag{1.8}$$

Theorem 1.2 (A streamlined framework)**.** *For n, k satisfying* (1.5)*, the argument of Section 4 gives* (1.2) *for*

$$p > \max\{p_b(n,k), p_N(n,k)\}. \tag{1.9}$$

Consequently, the restriction estimate (1.2) *holds for*

$$p > 2 + \frac{2.58631445\ldots}{n} + O\left(\frac{1}{n^2}\right)$$

For $n = 9$, this framework yields the best-known restriction exponents $p > 2150/929$.

1.2. **Structure of the article.** Section 2 fixes the geometric, wave-packet, and broad-measure conventions. Section 3 proves the scalar interpolation formulas and closes the Wang–Wu-type induction. Section 4 develops the transverse incidence estimate, the mixed broad bound, and the parameter-dependent narrow induction. The four-dimensional planebrush estimates are motivation only and are not used as unconditional inputs in any dimension.

1.3. **Notations and Conventions.** We write $A \lesssim B$ when $A \le CB$ with a constant depending only on fixed parameters, and display any dependence that is relevant to an induction. At a geometric scale δ, the notation $A \lessapprox B$ allows $C_\epsilon \delta^{-\epsilon}$ for every $\epsilon > 0$, together with any explicitly mentioned $\delta^{-C\epsilon_1}$ loss in a two-ends hypothesis. At a physical scale r the analogous convention is used. The notation must not

absorb a fixed negative power of a dynamic density or angle. Logarithmic losses are written separately when broad parameters are involved. A term $\mathrm{RapDec}(r)$ is bounded by $C_N r^{-N}$ for any prescribed N after the packet-tail cutoff has been chosen.

A unit δ-tube is a fixed constant enlargement of a length-one line segment by a transverse radius δ. Tube directions are unoriented; in the paraboloid chart we use representatives with positive last coordinate. A family is m-parallel and direction-separated if its directions lie in δ-separated angular classes and each class contains at most m tubes. Fixed direction colorings allow the usual buffered angular caps.

2. Preliminaries

2.1. **Two-ends shadings.** A shading $Y(T)$ is a union of cells of a fixed δ-grid contained in a fixed enlargement of T. Let

$$E_Y = \bigcup_{T\in\mathbb{T}} Y(T), \qquad \mu_Y(x) = \sum_{T\in\mathbb{T}} \mathbf{1}_{Y(T)}(x).$$

The shading is λ-dense if $|Y(T)| \asymp \lambda|T|$ for every retained tube. Later, for a refined subfamily, λ may instead denote the explicitly defined *average* density; this does not impose a new two-ends hypothesis on that subfamily.

Definition 2.1 (Two ends)**.** For $0 < \epsilon_2 < \epsilon_1 < 1$, a shading is (ϵ_1, ϵ_2)-two-ends if

$$|Y(T) \cap B(x, \delta^{\epsilon_1})| \le C\delta^{\epsilon_2}|Y(T)|$$

for every T and every such ball. Harmless fixed dilations and fixed constants are incorporated by decreasing ϵ_2 slightly. The constants in the geometric inputs may depend on ϵ_2.

The Wolff axiom with error m means that, for every box of dimensions $1 \times s_1 \times \cdots \times s_{n-1}$, with $\delta \le s_j \le 1$, the number of tubes contained in a fixed dilation of the box is at most

$$Cm\, s_1 \cdots s_{n-1}\delta^{-(n-1)}. \tag{2.1}$$

An m-parallel direction-separated family has this property: a contained tube must point in an angular region having the indicated product size, and each allowable direction contributes at most m tubes. In particular,

$$\delta^{n-1}\#\mathbb{T} \lesssim m. \tag{2.2}$$

2.2. **Geometric inputs and their precise roles.** The following statements are used with their small-loss versions. They are external inputs, not assertions of new incidence geometry.

Theorem 2.2 (General two-ends hairbrush; Gao–Wu–Xi)**.** *Suppose that $\mathbb{T}$ satisfies* (2.1) *and has a λ-dense, (ϵ_1, ϵ_2)-two-ends shading. Then* [GWX, Lemma 4.1] *gives*

$$|E_Y| \gtrsim_\epsilon \delta^{\epsilon + C\epsilon_1} m^{-\frac{n}{2n-2}} \lambda^{7/4} \delta^{(n-2)/2} (\delta^{n-1}\#\mathbb{T})^{\frac{n}{2n-2}}. \tag{2.3}$$

There is a good set $E_H \subset E_Y$ with $|E_Y \setminus E_H| \le \delta^{\epsilon_1}|E_Y|$ on which

$$\mu_Y \lesssim_\epsilon \delta^{-\epsilon - C\epsilon_1} m^{\frac{n}{2n-2}} \lambda^{-3/4} \delta^{-(n-2)/2} (\delta^{n-1}\#\mathbb{T})^{\frac{n-2}{2n-2}}. \tag{2.4}$$

Consequently, for an m-parallel direction-separated family,

$$\mu_Y \lessapprox m\lambda^{-3/4}\delta^{-(n-2)/2} \quad \text{on } E_H. \tag{2.5}$$

The exponent $(n-2)/(2n-2)$ in (2.4) is retained in the statement. No stronger version from a transverse subcase is applied to a general descendant family.

Theorem 2.3 (Two scalar inputs from Wang–Wu)**.** *For an m-parallel direction-separated, λ-dense two-ends family, the bush bound in* [WW, Corollary 3.4] *gives a good-set estimate*

$$\mu_Y \lessapprox m\delta^{-(n-1)/2}. \tag{2.6}$$

For $\lambda \geq \delta^{1/4}$, the Katz–Tao input as formulated in [WW, Theorem 3.5 and Corollary 3.7] *gives*

$$\mu_Y \lessapprox m\lambda^{-(2n+7)/7}\delta^{-3(n-1)/7}. \tag{2.7}$$

Both good sets have exceptional volume bounded by a positive small power of δ times $|E_Y|$. The bound (2.6) *is available at every density; only* (2.7) *has the displayed density restriction.*

The Katz–Tao theorem used here originates in [KT]. We use the precise density range in the formulation above, not an unstated extension of it.

Theorem 2.4 (Available Kakeya maximal input)**.** *Let $d = d_Z(n)$ as in* (**??**)*. If $\mathbb{T}$ is direction-separated, then*

$$\left\| \sum_{T\in\mathbb{T}} \mathbf{1}_T \right\|_{L^{d/(d-1)}} \lesssim_\epsilon \delta^{1-n/d-\epsilon} \left(\delta^{n-1}\#\mathbb{T}\right)^{(d-1)/d}. \tag{2.8}$$

This follows from [HRZ, Theorem 1.2] *or* [Zahl, Theorem 1.5].

Theorem 2.5 (Direction-separated multilinear Kakeya; Zahl)**.** *Let $2 \leq k \leq n$ and $d = d_{n,k}$. For a direction-separated family of unit δ-tubes,* [Zahl, Theorem 1.3] *gives*

$$\begin{aligned} &\left\| \left(\sum_{T_1,\dots,T_k\in\mathbb{T}} \mathbf{1}_{T_1}\cdots\mathbf{1}_{T_k} |v(T_1)\wedge\cdots\wedge v(T_k)|^{k/d} \right)^{1/k} \right\|_{d/(d-1)} \\ &\qquad \lesssim_\epsilon \delta^{1-n/d-\epsilon} (\delta^{n-1}\#\mathbb{T})^{\frac{n(d-1)}{(n-1)d}}. \end{aligned} \tag{2.9}$$

The exponent of the total tube volume in (2.9) differs from that in (2.8). Section 4 uses exactly (2.9), without invoking the multilinear-to-linear Kakeya reduction.

2.3. **Broad measures and localization conventions.** For the broad part of the paper, fix an outer radius R, and a dyadic $K \asymp R^\kappa$, where $\kappa > 0$ will be small. Cover the frequency ball by finitely overlapping caps τ of radius K^{-1}. The bounded overlap is independent of K. We permit sector-indexed inputs $\boldsymbol{f} = (f_\tau)_\tau$, with the support of each f_τ contained in a small buffer of τ. This formulation is useful after wave-packet selection. Define

$$\|\boldsymbol{f}\|_2^2 = \sum_\tau \|f_\tau\|_2^2, \qquad \|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)}^2 = \frac{1}{|\theta|}\sum_\tau \int_\theta |f_\tau|^2. \tag{2.10}$$

Partition physical space into small cubes b of side comparable to K^2. For a vector of functions $\boldsymbol{H} = (H_\tau)_\tau$ set

$$\mu_{k,A,p}(\boldsymbol{H}; b) = \inf_{V_1,\dots,V_A} \max_{\substack{\tau:\ \mathrm{angle}(v(2\tau),V_a)>K^{-1} \\ \text{for every } a}} \int_b |H_\tau|^p, \tag{2.11}$$

where each V_a is a $(k-1)$-dimensional linear subspace and the angle is an infimum over the sector. The maximum of an empty set is zero. On unions of these cubes define

$$\|\boldsymbol{H}\|^p_{\mathrm{BL}^p_{k,A,K}(U)} = \sum_{b\subset U} \mu_{k,A,p}(\boldsymbol{H}; b). \tag{2.12}$$

Bounded changes in cube sizes, cap buffers and the angle constant are harmless. One may use a finite collection of shifted grids at boundaries. All larger cubes used below are unions of the small cubes.

At intermediate radius r, sector-support buffers are of size $Cr^{-1/2}$. If r is replaced by $r^{1-\alpha}$, their allowed size increases by a power of r, and absorbs the support enlargement in wave-packet projection. All these buffers lie inside 2τ because the iteration stops long before r approaches a power of K of the relevant size.

Lemma 2.6 (Subadditivity with parameter loss)**.** *If $\boldsymbol{H} = \sum_{j=1}^{L} \boldsymbol{H}_j$ and $\sum_j A_j \le A$, then*

$$\mu_{k,A,p}(\boldsymbol{H}; b)^{1/p} \le \sum_{j=1}^{L} \mu_{k,A_j,p}(\boldsymbol{H}_j; b)^{1/p}. \tag{2.13}$$

The analogous inequality holds for the broad norms on U.

Proof. Choose almost-minimizing exceptional planes separately for each j, and take their union. Every cap outside that union is outside each separate collection. Apply Minkowski on b, take the maximum, and then the infimum. Sum in ℓ^p_b. □

Lemma 2.7 (Extraction of genuinely transverse sectors)**.** *Let $A \ge 1$. At the cost of $K^{C_{n,k}}$, a broad measure can be bounded by a sum over finitely many k-tuples of sector subfamilies with the following property: every choice of their directions satisfies*

$$|v_1 \wedge \cdots \wedge v_k| \gtrsim K^{-C_{n,k}}. \tag{2.14}$$

After grouping by the tuple, and summing over a union X of larger cubes, its contribution is at most

$$K^{C_{n,k}} \left(\prod_{i=1}^{k} \int_X |G_i|^p \right)^{1/k}, \tag{2.15}$$

where G_i is the sum in the ith extracted sector.

Proof. For one small cube, denote the pth root of its broad measure by h. Given any $(k-1)$-plane, repeat it A times in (2.11); some cap outside it has $L^p(b)$ norm at least h. Subdivide caps into sectors of radius K^{-L} with fixed $L > 2k + 10$. By Minkowski, some subsector has norm at least $K^{-C}h$. Choose one such subsector, then choose an exceptional plane containing its center direction, then one containing the span of the first two chosen directions, and continue. Each successive center is at distance at least a constant times K^{-1} from the preceding span. Their wedge is at least a constant times $K^{-(k-1)}$. Since the subsectors have radius K^{-L}, this lower bound persists for all directions in them. Thus h^p is at most K^C times the minimum of the k subsector integrals.

There are at most K^C possible tuples. Sum their contributions. For a fixed tuple, if Q is a larger cube,

$$\sum_{b\subset Q}\min_i\int_b|G_i|^p\le\left(\prod_i\int_Q|G_i|^p\right)^{1/k}.$$

A second application of Hölder in the Q sum proves (2.15). For the application, we retain the intermediate functional

$$\mathfrak{b}_p(\boldsymbol{G};X)=\sum_{b\subset X}\min_i\int_b|G_i|^p$$

when partitioning X by the separate multiplicities. This functional is additive in disjoint unions of the small cubes, so one multiplicity bin carries a polylogarithmic fraction of it. Only after this selection do we use the upper bound by the geometric mean of the integrals over the selected region. A geometric mean of integrals over the original region alone would not justify this selection: the different integrals could be concentrated on disjoint sets. For wave-packet sums, subsectors are formed by grouping packet directions, not by applying new frequency multipliers to the packets. Their norm uniformity is therefore unchanged. The condition $r^{-1/2}\ll K^{-L}$ ensures support buffers remain transverse. □

2.4. **Wave packets, average energy, and refined decoupling.** Put $s=s_n=2(n+1)/(n-1)$. At radius r, use frequency caps θ of radius $r^{-1/2}$ and packets f_T supported in fixed dilates of these caps. In physical space the packets are negligible outside tubes of radius $r^{1/2+\varepsilon_0}$ and length comparable to r, with bounded slopes relative to the x_n axis. Spatial packet centers for a fixed direction form an $r^{1/2}$ lattice. We retain losses $r^{C_n\varepsilon_0}$ from tube enlargement and local overlap. Rapidly decreasing errors can be made $O(r^{-M})\|\boldsymbol{f}\|_2$ for any fixed M, after excluding an equally negligible packet tail.

The following properties hold uniformly for a packet subfamily and, with energy multiplicity, for a collection of such subfamilies:

$$\left\|\sum_{T\in\mathcal{S}}f_T\right\|_2^2\lesssim\sum_{T\in\mathcal{S}}\|f_T\|_2^2,\qquad\sum_T\|f_T\|_2^2\lesssim\|\boldsymbol{f}\|_2^2,\tag{2.16}$$

$$\max_{\theta:\,t^{-1/2}\text{-cap}}\left\|\sum_{T\in\mathcal{S}}f_T\right\|_{L^2_{\mathrm{avg}}(\theta)}\lesssim\max_{\theta:\,r^{-1/2}\text{-cap}}\|\boldsymbol{f}\|_{L^2_{\mathrm{avg}}(\theta)},\qquad t\le r.\tag{2.17}$$

In (2.17) the packet sum keeps its original sector labels, and the average radius $t^{-1/2}$ is no smaller than $r^{-1/2}$.

For completeness, one may choose localized Fourier-series packets on each frequency cap. Their coefficient square sum is controlled by Parseval, and the Gram matrix of the modulated smooth cap cutoffs has rapid off-diagonal decay in the spatial-lattice index. Schur's test therefore gives the upper almost-orthogonality estimate for every subfamily. The same construction gives the coefficient-energy bound for the original input on a fixed enlargement of each cap. Finite frequency overlap permits summation over caps and sector labels. For an average on a ball ω with radius at least $r^{-1/2}$, only frequency caps within $C\omega$ contribute. Apply these energy estimates to those caps and use the average bound for the original input on

$C\omega$. Translation to a new physical center multiplies the input by a unit-modulus phase, and changes neither this estimate nor the frequency support.

After dyadic selection we may assume

$$\|f_T\|_2 \asymp a, \qquad \#\{T : \theta(T) = \theta\} \asymp m \tag{2.18}$$

for the original retained family and each active fine direction. Further selections need only have at most Cm packets in each direction. One always retains

$$a^2 m \lesssim r^{-(n-1)/2} \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^2_{L^2_{\text{avg}}(\theta)}, \qquad a^2 \#\mathbb{T} \lesssim \|\boldsymbol{f}\|_2^2. \tag{2.19}$$

Only logarithmically many amplitude and directional-count bins are needed: discarded lower bins contribute a rapidly decreasing error. The relevant broad parameter decreases by a polylogarithmic factor, by Lemma 2.6.

Incidence is defined using one fixed sufficiently large constant dilation of each packet tube. The associated shading consists of whole grid cells and lies in a further fixed dilation. Local orthogonality and refined decoupling only require that these counts dominate the genuinely contributing packets. This common convention is maintained through all refinements, and all fixed tube dilations are allowed in the geometric estimates.

Let X be a union of cubes of side $r^{1/2+\varepsilon_0}$. Suppose that on a region under consideration the time-truncated sum in sector i agrees with the complete extension of a fixed packet subfamily $\mathbb{T}_i$, up to a rapidly decreasing error. Let $N_i = \#\mathbb{T}_i$, and let $\mu_i(Q)$ count the active tubes meeting an enlarged version of Q. Define the average density λ_i by

$$\sum_{T \in \mathbb{T}_i} |Y_i(T)| = \lambda_i N_i |T|, \tag{2.20}$$

with the usual bounded-volume-equivalence convention for tubes and grid shadings. Then local frequency orthogonality gives

$$\int_X |G_i|^2 \lesssim r^{C\varepsilon_0} \lambda_i r\, a^2 N_i. \tag{2.21}$$

Indeed, on one Q the L^2 integral is at most $r^{1/2+C\varepsilon_0} \sum_{T:\, Q\sim T} \|f_T\|_2^2$. Sum over Q and use (2.20). Since all packet energies are comparable, only the *average* incidence density is needed. No pointwise lower bound for a packet on its tube is used.

If $\mu_i(Q) \lesssim \mu_i$ throughout X, elliptic refined decoupling [GIOW, Theorem 4.2], in the form [WW, Theorem 4.4], implies

$$\begin{aligned} \int_X |G_i|^s &\lesssim_\varepsilon r^{\varepsilon + C\varepsilon_0} \mu_i^{2/(n-1)} a^s N_i \\ &\lesssim_\varepsilon r^{-1+\varepsilon+C\varepsilon_0} (\mu_i/m)^{2/(n-1)} a^2 N_i \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{s-2}_{L^2_{\text{avg}}(\theta)}. \end{aligned} \tag{2.22}$$

To see the packet-size bound in the first inequality, use $\|Ef_T\|_\infty \lesssim r^{-(n-1)/4} a$ and $\|Ef_T\|^2_{L^2(w_{B_r})} \lesssim ra^2$. Their interpolation gives $\|Ef_T\|^s_{L^s(w_{B_r})} \lesssim a^s$. If the packet L^s norms are not comparable, decompose them dyadically in the refined decoupling inequality; this costs only logarithms and does not change the geometric multiplicity upper bound. The second inequality in (2.22) follows from (2.19).

3. Refining the Wang–Wu framework

3.1. **The scalar analytic inputs.** We first calculate on a single uniformized, two-ends piece. The scale is r, $\delta = r^{-1/2}$, the packet energy is $\|f_T\|_2 \asymp a$, and the initial number of packets per active direction is m. Write

$$s = \frac{2(n+1)}{n-1}, \qquad I_t = \int_X |G|^t.$$

On a fixed common time slab, G is the extension of a fixed packet subfamily on the region being integrated. Its shading is a subset of an ancestor shading of density λ. Consequently (2.21) and (2.22) imply

$$\begin{aligned} I_2 &\lessapprox r\lambda \|f\|_2^2, \\ I_s &\lessapprox r^{-1}(\mu/m)^{2/(n-1)} \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{s-2}. \end{aligned} \tag{3.1}$$

Using the *ancestor* density in the first inequality is legitimate since the refined average density is no larger. The good-set bounds on the ancestor multiplicity also remain upper bounds for the refined multiplicity.

The bush, hairbrush, Katz–Tao, and maximal inputs yield the following alternatives:

$$I_s \lessapprox r^{-1/2} \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{s-2}, \tag{3.2}$$

$$I_s \lessapprox r^{-\frac{n}{2(n-1)}} \lambda^{-\frac{3}{2(n-1)}} \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{s-2}, \tag{3.3}$$

$$I_s \lessapprox r^{-4/7} \lambda^{-\frac{2(2n+7)}{7(n-1)}} \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{s-2}, \qquad \lambda \geq r^{-1/8}, \tag{3.4}$$

$$I_s \lessapprox r^{-\frac{d-1}{n-1}} \lambda^{-\frac{2(d-1)}{n-1}} \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{s-2}. \tag{3.5}$$

In (3.5), d may be any parameter for which (2.8) is available. The maximal-to-shading conversion is proved next; the other alternatives follow directly from the geometric inputs in Section 2.2.

Lemma 3.1 (From a maximal estimate to a good-set bound)**.** *Assume* (2.8) *for* $2 < d \leq n$. *Let* $\mathbb{T}$ *be* m*-parallel and direction-separated and let its shading have average density* λ. *For every* $\sigma > 0$ *there is a set* $E_M \subset E_Y$ *with* $|E_Y \setminus E_M| \leq \delta^\sigma |E_Y|$ *such that*

$$\mu_Y(x) \lesssim_{\epsilon,\sigma} \delta^{-\epsilon-\sigma} m \lambda^{-(d-1)} \delta^{-(n-d)}, \qquad x \in E_M. \tag{3.6}$$

The bound is valid without a two-ends assumption.

Proof. Set $W = \sum_T |T|$ and $r_d = d/(d-1)$. Split the tubes into at most Cm direction-separated families. Minkowski, followed by Hölder in the family index, gives

$$\left\| \sum_T \mathbf{1}_T \right\|_{r_d} \lesssim_\epsilon m^{1/d} \delta^{1-n/d-\epsilon} W^{(d-1)/d}.$$

Since $\int \mu_Y \asymp \lambda W$, Hölder on E_Y implies

$$\lambda W \lesssim |E_Y|^{1/d} m^{1/d} \delta^{1-n/d-\epsilon} W^{(d-1)/d}.$$

After changing the arbitrary small exponent,

$$|E_Y| \gtrsim_\epsilon \delta^\epsilon m^{-1} \lambda^d \delta^{n-d} W. \tag{3.7}$$

It follows that the average of μ_Y on E_Y is at most $C_\epsilon \delta^{-\epsilon} m \lambda^{-(d-1)} \delta^{-(n-d)}$. Markov's inequality with a factor $\delta^{-\sigma}$ proves (3.6). □

The good sets needed for any fixed interpolation are intersected *before* the last physical refinement. Their exceptional volumes are each a small power of δ. The reduction below makes the retained L^p mass per mesoscopic cube comparable, so this intersection still carries the required integral. Two unrelated good sets cannot simply be used as though their estimates held on the same integration set.

3.2. **Hairbrush alone and hairbrush plus a maximal estimate.** For the hairbrush bound (3.3), choose

$$t = \frac{2(n-1)}{3n-2}.$$

Then the r-power in $I_2^{1-t} I_s^t$ is zero, and the remaining density factor is

$$\lambda^{(n-3)/(3n-2)} \le 1.$$

The corresponding threshold is $P_{\mathrm{H}}(n) = 2 + 8/(3n-2)$.

For the combined maximal and hairbrush estimate, let

$$D_d = (4n-5)(d-1) + (n-1)(n-3),$$

and set

$$u = \frac{(2n-3)(d-1)}{D_d}, \qquad v = \frac{2(n-1)(d-1)}{D_d}, \qquad w = \frac{(n-1)(n-3)}{D_d}. \tag{3.8}$$

These weights are nonnegative and sum to one. The powers of r and λ cancel because

$$u - \frac{n}{2(n-1)} v - \frac{d-1}{n-1} w = 0, \tag{3.9}$$

$$u - \frac{3}{2(n-1)} v - \frac{2(d-1)}{n-1} w = 0. \tag{3.10}$$

Apply Hölder to $I_2^u I_s^{v+w}$, using (3.3) for the v factor and (3.5) for the w factor. The result is

$$I_p \lessapprox \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}^{p-2}, \qquad p = 2u + s(v+w) = \mathfrak{P}(n,d). \tag{3.11}$$

Taking $d = d_Z(n)$ proves the uniformized two-ends piece needed for $P_{\mathrm{M}}(n)$. Both geometric inputs are available at every density, so no density range is omitted.

For instance, when $n = 5$ and $d_Z(5) = 18/5$, the weights are

$$(u, v, w) = \frac{1}{235}(91, 104, 40), \qquad P_{\mathrm{M}}(5) = \frac{614}{235}.$$

The cancellations reduce to $91 - 65 - 26 = 0$ and $91 - 39 - 52 = 0$. These identities illustrate why a better scale exponent and a worse density exponent can still be useful when combined with the hairbrush bound.

3.3. **Hairbrush plus Katz–Tao.** Let $n \geq 9$ and write $D_0 = 11n^2 + 10n - 3$. In the large-density range $\lambda \geq r^{-1/8}$, use

$$u = \frac{4n^2 + 2n + 12}{D_0}, \qquad v = \frac{36(n-1)}{D_0}, \qquad w = \frac{7(n-1)(n-3)}{D_0}. \tag{3.12}$$

The identities

$$\begin{aligned} u + v + w &= 1, \\ u - \frac{n}{2(n-1)}v - \frac{4}{7}w &= 0, \\ u - \frac{3}{2(n-1)}v - \frac{2(2n+7)}{7(n-1)}w &= 0 \end{aligned}$$

show that (3.1), (3.3), and (3.4) give

$$I_p \lessapprox \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}^{p-2}, \qquad p = 2 + \frac{4}{n-1}(v+w) = P_{\mathrm{K}}(n). \tag{3.13}$$

For $\lambda \leq r^{-1/8}$, discard the unavailable Katz–Tao input and use only the hairbrush estimate, with total high-exponent weight

$$t = v + w = \frac{(n-1)(7n+15)}{D_0}.$$

The scale and density factor in the interpolation is

$$r^{\frac{(n-8)(n-3)}{2D_0}} \lambda^{\frac{(8n+7)(n-3)}{2D_0}} \leq r^{-\frac{71(n-3)}{16D_0}} \leq 1. \tag{3.14}$$

Thus the same output exponent controls every density. The original small-power losses, and any small shift of the availability boundary caused by tube enlargement, are absorbed by choosing the geometric parameters sufficiently small. The strict negative power in (3.14) also leaves room at this boundary.

3.4. **Closing the scalar two-ends induction.** For completeness we spell out the closure shared by the preceding calculations. This is the Wang–Wu reduction [WW, Section 5], arranged with common time intervals so that the restricted packet sums have a legal extension interpretation.

Fix a target loss $\eta > 0$ and a large outer radius R. Select positive parameters

$$n\theta \ll \eta, \qquad C_n\alpha \ll \eta, \qquad b_0 \ll \eta\alpha, \qquad \epsilon_0, \epsilon' \ll b_0.$$

For one of the exponents $p = P_{\mathrm{H}}(n), P_{\mathrm{M}}(n), P_{\mathrm{K}}(n)$ in its stated range, prove the mixed estimate

$$\int_{B_r} |Ef|^p \leq C_\eta R^\eta r^\eta \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}^{p-2}, \qquad R^\theta \leq r \leq R, \tag{3.15}$$

by descending through radii $r_1 = r^{1-\alpha}$. Start a sequence at any requested radius and stop at $r \leq R^\theta$. The number of steps depends on the small parameters but not on R. At the bottom, $r^n \|f\|_2^p \lesssim R^{n\theta}\|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}^{p-2}$ is sufficient.

At radius r, decompose into packet-energy bins and numbers of packets per active direction. There are only $O((\log r)^C)$ relevant bins, after negligible terms have been removed. The triangle inequality and a final Hölder inequality cost a polylogarithm. The estimates (2.16)–(2.19) permit their selected pieces to be measured by the original $\|f\|_2$ and $\max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}$.

Select a union X of mesoscopic cubes, of side $r^{1/2+\epsilon_0}$, on which the L^p mass is comparable. Partition time into common intervals of length r_1. For each tube, select a dyadic segment mass $\lambda|T|$ and a dyadic count β of active segments of that mass. Let $Y(T)$ be the union of the incidences with X in these active segments. Its density is

$$\lambda_0 \asymp \beta\lambda.$$

On one common time slab, the resulting sum agrees on X with the extension of a fixed packet subfamily, modulo rapidly decreasing tails. The spatial membership of a shading is used to ignore noncontributing tubes, not to assert that deletion of packets decreases the absolute value of their sum.

The triangle inequality over the finitely many density and count bins gives a selected truncated sum whose integral is a polylogarithmic fraction of the original one. One may retain a union $X_2 \subset X$ of at least $|X|/(\log r)^C$ mesoscopic cubes and make its selected integral per cube comparable. Here is the elementary selection principle being used. If $I_Q \asymp I$ and $I_Q \le L^C \sum_{j\le L} I_{Q,j}$, choose an index contributing at least $L^{-C-1}I$ on each cube; one index occurs on at least a $1/L$ fraction of the cubes, and a final dyadic selection makes its values comparable. All values have a polynomial scale range after negligible tails are discarded. This proves simultaneous mass and volume retention for these selections.

If $\beta \le r^{b_0}$, subdivide each active slab spatially into r_1-cubes. Denote the corresponding fixed packet sums by f_j. Each tube meets only $O_n(1)$ spatial cubes per active slab. The packet energy and average-energy bounds give

$$\begin{aligned} \sum_j \|f_j\|_2^2 &\lesssim r^{b_0+C_n\epsilon_0}\|f\|_2^2, \\ \max_{\theta:\, r_1^{-1/2}\text{-cap}} \|f_j\|_{L^2_{\text{avg}}(\theta)} &\lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}. \end{aligned} \tag{3.16}$$

Applying the induction hypothesis at r_1 leaves a relative factor

$$C(\log r)^C r^{-\eta\alpha+b_0+C_n\epsilon_0},$$

which is less than $1/2$ at sufficiently large scales.

If $\beta > r^{b_0}$, a ball of radius comparable to r_1 meets only $O_n(1)$ active time intervals. Equal segment masses therefore give the two-ends condition for $Y(T)$, with first parameter comparable to 2α and a positive second parameter smaller than $2b_0$. Apply all geometric good-set bounds required for the chosen interpolation to this ancestor family. Their union of exceptional sets has measure at most $r^{-c\alpha}|X|$. It cannot remove a fixed fraction of the selected integral on X_2, since the latter has comparable mass per whole grid cube and $|X_2| \gtrsim |X|/(\log r)^C$.

After retaining the common good set, fix one common slab and one r_1-cube at a cost of at most $r^{C_n\alpha}$. The packet subfamily on it is fixed, and its average shading density is at most $C\lambda_0$. Apply the appropriate interpolation from the previous three subsections with λ_0 in place of λ. The bounds remain valid after restricting the integration set and after passing to a packet subfamily, with energy measured by the original $\|f\|_2$ and $\max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}$. The result is bounded by

$$C_\eta r^{C_n\alpha+C_n\epsilon_0+\epsilon'}(\log r)^C \|f\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}^{p-2}.$$

The chosen hierarchy absorbs this into (3.15). This completes both cases and proves the mixed local bound at $r = R$ with an arbitrarily small total loss.

For $|f| \le \mathbf{1}_\Omega$, the mixed right side is at most $CR^{2\eta}|\Omega|$. Real interpolation gives the local diagonal estimate above the mixed exponent; equivalently one may use [WW, Lemma 4.7]. Finally choose a slightly smaller local exponent than the desired global exponent and apply the epsilon-removal lemma of Tao [Tao99], as recalled at the beginning of [WW, Section 4]. This proves Theorem 1.1.

Remark 3.2. A refined shading is not assumed to retain the two-ends condition. Only the ancestor shading is supplied to the geometric theorems. Its good-set multiplicity bound is inherited by subfamilies, and its density is an upper bound for their average densities. Conversely, an upper bound on the cardinality of a tube family does not imply a pointwise bound on the Fourier sum. The common time slabs and the packet localizations are needed for this reason.

4. A multilinear broad–narrow framework

Fix n, k as in (1.5). In this section

$$d = d_{n,k}, \qquad s = \frac{2(n+1)}{n-1}, \qquad D = (4n-5)(d-1) + (n-1)(n-3), \tag{4.1}$$

and p_b, p_N are given by (1.6)–(1.7). The goal is to preserve the value $d_{n,k}$ until after the Fourier broad estimate. All the transverse statements below retain their transversality hypotheses.

4.1. The local incidence parameter behind the planebrush motivation.

We first isolate a local estimate that explains the connection with the retained-multiplicity viewpoint. It is not an unconditional higher-dimensional planebrush theorem and is not an extra unproved input for the later argument.

Lemma 4.1 (A local incidence-retaining inequality)**.** *Let $\mathbb{T}_Q$ be a direction-separated family of unit δ-tubes and let E_Q be a union of δ-cells. Suppose the shaded multiplicity is comparable to $\mu > 0$ on E_Q, and on every cell at least $c\mu^k$ of the incident ordered k-tuples have wedge at least $\tau > 0$. Let $N_Q = \#\mathbb{T}_Q$ and let I_Q be the number of shaded tube–cell incidences. Put $\gamma = (d-1)/(n-1)$. If*

$$N_Q \lesssim \mu_\rho D_{\mathrm{desc}}, \qquad I_Q \gtrsim \mu_\rho M_{\mathrm{inc}},$$

then

$$\mu \lesssim_{\epsilon,c} \delta^{-(n-d)-\epsilon}\tau^{-1}\mu_\rho^{\gamma} D_{\mathrm{desc}}^{n\gamma} M_{\mathrm{inc}}^{-(d-1)}. \tag{4.2}$$

Proof. The transverse tuples and (2.9) imply

$$\mu^{d/(d-1)}\tau^{1/(d-1)}|E_Q| \lesssim_{\epsilon,c} \delta^{-(n-d)/(d-1)-\epsilon}(\delta^{n-1}N_Q)^{n/(n-1)}.$$

Substitute $|E_Q| \asymp \delta^n I_Q/\mu$ and raise to the power $d-1$. This gives

$$\mu \lesssim_{\epsilon,c} \delta^{-(n-d)-\epsilon}\tau^{-1}\left(\frac{N_Q^{n/(n-1)}}{I_Q}\right)^{d-1}.$$

The stated incidence and cardinality hypotheses finish the proof, since $n\gamma-(d-1) = \gamma$. □

For example, if a coarse/fine organization provides

$$D_{\mathrm{desc}} \asymp (\rho/\delta)^{n-1}X/X_\rho, \qquad M_{\mathrm{inc}} \gtrapprox (\rho/\delta)D_{\mathrm{desc}}\chi^{-1},$$

then (4.2) becomes

$$\mu \lessapprox \delta^{-(n-d)}\tau^{-1}\chi^{d-1}\mu_\rho^{\gamma}(X/X_\rho)^{\gamma}. \tag{4.3}$$

Here $X = \delta^{n-1}\#\mathbb{T}$ and $X_\rho = \rho^{n-1}\#\mathbb{T}_\rho$ are normalized tube counts, and χ is the coarse-to-fine shading-density ratio. This calculation requires the displayed hypotheses; it does not establish them for an arbitrary configuration.

A direct high-dimensional planebrush proof would have to complement (4.3) in all angularly concentrated configurations and remove the residual parameters ρ, τ, χ. We do not undertake that task here. Instead, broadness supplies transverse directions, the next lemma deals with their unequal populations, and narrow decoupling handles concentration on the Fourier side. This is the precise sense in which the present framework is a multilinear substitute for the desired geometric argument.

4.2. **A geometric lemma allowing unequal transverse populations.**

Lemma 4.2. *Let $\mathbb{T}_1, \dots, \mathbb{T}_k$ be disjoint angular families of unit δ-tubes, and suppose every choice $T_i \in \mathbb{T}_i$ has wedge at least $\nu > 0$. There are at most m tubes in each δ-separated direction class. Let E be a union of $O(\delta^{-n})$ grid cells of side δ. Assume the shaded multiplicity in family i is comparable to a positive constant μ_i on every cell of E. Set*

$$N_i = \#\mathbb{T}_i, \qquad N = \sum_i N_i, \qquad Z = \frac{\delta^{n-1}N}{m},$$

$$\zeta = \frac{(N_1 \cdots N_k)^{1/k}}{N}, \qquad \Lambda = (\lambda_1 \cdots \lambda_k)^{1/k}, \qquad \mu = (\mu_1 \cdots \mu_k)^{1/k}.$$

where $\mu_i|E| \asymp \lambda_i \delta^{n-1} N_i$ defines the average densities. If $\min_i \mu_i/m \geq C_{n,k}\log(2/\delta)$, then

$$\frac{\mu}{m} \lesssim_\varepsilon \nu^{-1}\delta^{-(n-d)-\varepsilon}\Lambda^{-(d-1)}\zeta^{-(d-1)}Z^{(d-1)/(n-1)}. \tag{4.4}$$

Proof. Pad each direction class with dummy entries to have m slots, and choose one slot uniformly and independently in each class. For each fixed cell and family i, the sampled shaded multiplicity is a sum of independent Bernoulli variables, one for each direction class, of total expectation comparable to μ_i/m. Chernoff's inequality and a union bound over the polynomially many cells show that all these sampled multiplicities are at least $c\mu_i/m$ simultaneously, with probability greater than $3/4$, when the displayed logarithmic threshold is large enough. The expected total number of sampled tubes is N/m; Markov's inequality makes it at most $8N/m$ with probability at least $7/8$. Thus a realization with both properties exists. A bounded direction coloring handles adjacent direction cells when necessary.

The chosen realization is direction-separated. Apply (2.9) to its complete tubes, using their shaded incidences only for a lower bound on the integrand on E. Raise the result to $d/(d-1)$:

$$\nu^{1/(d-1)}(\mu/m)^{d/(d-1)}|E| \lesssim_\varepsilon \delta^{-(n-d)/(d-1)-\varepsilon}Z^{n/(n-1)}. \tag{4.5}$$

Taking geometric means of the incidence identities gives

$$\mu|E| \asymp \Lambda\zeta\delta^{n-1}N = \Lambda\zeta m Z.$$

Substitute this into (4.5), cancel, and raise to the power $d-1$. This proves (4.4), after renaming the arbitrary small loss. □

Remark 4.3. The lemma does not require comparable N_i or comparable μ_i. It makes no two-ends assumption on the refined shadings. These two features are essential in the application. For enlarged wave-packet tubes, grouping fine directions at the enlarged angular scale replaces m by at most $r^{C_n\varepsilon_0}m$; applying the lemma and translating back introduces only $r^{C_{n,k}\varepsilon_0}$ factors. The low-multiplicity alternative is treated directly in the next section.

4.3. **The transverse analytic branch.** Assume we have reached a common region X and k transverse packet families as in Sections 2.4–4.2. We also assume a good-set hairbrush upper bound for the *ancestor shading*:

$$\mu_i \le \mu_{\text{ancestor}} \lesssim r^e m\lambda_0^{-3/4} r^{(n-2)/4}, \tag{4.6}$$

where $e > 0$ is arbitrarily small apart from a fixed dimensional multiple of the two-ends scale parameter. Every refined shading has average density at most $C\lambda_0$. In particular $\Lambda \lesssim \lambda_0$, and (4.6) remains valid with $\lambda_0^{-3/4}$ replaced by $C\Lambda^{-3/4}$.

Here $a^2N \lesssim \|\boldsymbol{f}\|_2^2$. The numbers Z and ζ are bounded above by dimensional constants and by 1, respectively. The small enlargement errors in Z can instead be included in r^e. We estimate the local minimum functional from Lemma 2.7; an upper bound by a geometric mean of total integrals is invoked only on the selected common region.

If some $\mu_i/m \lesssim r^{C\varepsilon_0}\log r$, use this family's L^2 bound with density at most one and its refined decoupling bound. Interpolation with $t = (p_b-2)/(s-2) = 1-u$ gives a factor r^{1-2t+e}. Here

$$u = \frac{(2n-3)(d-1)}{D} < \frac{1}{2},$$

so the exponent without e is strictly negative. The local minimum functional is at most the integral of this one family over the selected region, hence this case satisfies the desired bound. We do not claim to control the geometric mean of all sector integrals by the low-multiplicity family.

In the remaining case apply Lemma 4.2. Put

$$\mathcal{I}_q = \left(\prod_{i=1}^{k}\int_X |G_i|^q\right)^{1/k}.$$

Taking geometric means of (2.21), and then of the two alternatives in (2.22), gives

$$\mathcal{I}_2 \lesssim r^e\, r\Lambda\zeta a^2N, \tag{4.7}$$

$$\mathcal{I}_s \lesssim r^e\, r^{-n/(2(n-1))}\Lambda^{-3/(2(n-1))}\zeta a^2N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\text{avg}}(\theta)}^{s-2}, \tag{4.8}$$

$$\begin{aligned}\mathcal{I}_s \lesssim{}& K^C r^e\, r^{-(d-1)/(n-1)}\Lambda^{-2(d-1)/(n-1)}\zeta^{1-2(d-1)/(n-1)}Z^{2(d-1)/(n-1)^2}\\ &\times a^2N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\text{avg}}(\theta)}^{s-2}.\end{aligned} \tag{4.9}$$

The polynomial K loss includes the wedge bound of Lemma 2.7.

Choose

$$u = \frac{(2n-3)(d-1)}{D}, \qquad v = \frac{2(n-1)(d-1)}{D}, \qquad w = \frac{(n-1)(n-3)}{D}. \tag{4.10}$$

These weights are nonnegative, sum to one, and satisfy

$$u - \frac{n}{2(n-1)}v - \frac{d-1}{n-1}w = 0, \tag{4.11}$$

$$u - \frac{3}{2(n-1)}v - \frac{2(d-1)}{n-1}w = 0. \tag{4.12}$$

Also $2u + s(v+w) = p_b$. Hölder and the two upper bounds for $\mathcal{I}_s$ yield

$$\begin{aligned} \mathcal{I}_{p_b} &\lesssim K^C r^e \zeta^\vartheta Z^\psi a^2 N \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\text{avg}}(\theta)} \\ &\lesssim K^C r^e \|\boldsymbol{f}\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\text{avg}}(\theta)}, \end{aligned} \tag{4.13}$$

where

$$\vartheta = \frac{(2n+1)(d-1)+(n-1)(n-3)}{D} > 0, \qquad \psi = \frac{2(d-1)(n-3)}{(n-1)D} \geq 0. \tag{4.14}$$

This is the precise reason the imbalance in transverse tube populations does not cost a power of the scale. Dropping ζ earlier would not be valid.

4.4. **The broad two-ends induction.** We now supply the scale bookkeeping needed to use the preceding branch for all inputs. This part uses broad measures from (2.11), not a presumed multilinear triangle inequality. The local minimum functional retained in Lemma 2.7 is used until the physical multiplicity bins have been selected. The use of a fixed outer scale and a growing broadness parameter has a related precedent in Demeter–Wu, Proposition 4.13 [DW]; the argument here is given for the present k-broad measures.

Proposition 4.4. *Fix n, k as in Theorem 1.2. Given $\eta > 0$, one can choose $\kappa > 0$ sufficiently small and then $a_0 > 0$ arbitrarily small such that, with $K \asymp R^\kappa$, the following holds:*

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_R)} \lesssim_\eta R^{2\eta} \|\boldsymbol{f}\|_2^2 \max_{\theta:\, R^{-1/2}\text{-}cap} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\text{avg}}(\theta)}, \qquad A \geq R^{a_0}. \tag{4.15}$$

All constants are independent of R and A. For canonical sector decompositions, $\|\boldsymbol{f}\|_2 \lesssim \|f\|_2$.

Proof. **Parameter order.** Choose a stopping exponent $\theta > 0$ with $n\theta < \eta/4$. Choose a time-shortening exponent $\alpha > 0$ small enough that every geometric loss $r^{C_{n,k}\alpha}$ below is bounded by $r^{\eta/4}$. Choose $b_0 > 0$ with $2b_0 < \eta\alpha/4$. Take the wave-packet enlargement exponent and the arbitrary losses in (2.9) and refined decoupling much smaller than b_0 and η. Choose κ so small that the extraction loss $K^{C_{n,k}} \leq R^{\eta/4}$ and $K^{2L} \ll R^\theta$. Finally take $a_0 > 0$ as small as desired, but fixed and positive. Increase the lower threshold for R so that every polylogarithmic loss is dominated by the positive powers used below.

For $r \in [R^\theta, R]$ we prove

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \leq C_\eta R^\eta r^\eta \|\boldsymbol{f}\|_2^2 \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|^{p_b-2}_{L^2_{\text{avg}}(\theta)}, \qquad A \geq r^{a_0}. \tag{4.16}$$

Use a discrete sequence with successive radii r and $r_1 = r^{1-\alpha}$, rounded dyadically. Its length down to R^θ is bounded in terms of η alone. For an intermediate requested radius, start the descending sequence at that radius. We do not cover a ball by only boundedly many balls of the strictly smaller power scale.

Base case. For $r \lesssim R^\theta$, the trivial bound is $Cr^n\|\boldsymbol{f}\|_2^{p_b}$. Since $\|\boldsymbol{f}\|_2 \lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\text{avg}}(\theta)}$ and $r^n \le CR^{\eta/4}$, this is absorbed by (4.16).

Packet selection and broad-mass regularization. At scale r, discard the rapidly decreasing packet tail and select common energy and per-direction-count bins as in (2.18). Lemma 2.6 costs a factor $(\log r)^C$ in the broad parameter and in the norm. Select a union X of $r^{1/2+\varepsilon_0}$-cubes on which the broad mass per cube is comparable and which carries the original broad norm up to logarithms.

All subsequent pigeonholing involving the density and the number of active time intervals can be made to preserve both a polylogarithmic fraction of the number of these cubes and the required broad integral. To verify this, if $I_Q \asymp I$ and $I_Q \le L^C \sum_{j=1}^L I_{Q,j}$, first choose $j(Q)$ with $I_{Q,j(Q)} \ge L^{-C-1} I_Q$, then a value of j occurring on at least $1/L$ of the cubes, and then a dyadic size of $I_{Q,j}$. The relevant sizes have polynomial range in r, after discarding negligible terms. This is an elementary finite pigeonholing argument, not a claim that an arbitrary refinement preserves the geometry.

Common time intervals. Partition the x_n axis into intervals of length $r_1 = r^{1-\alpha}$, aligned with the physical grid. For every tube, define its segments using these *same* time intervals. Select a dyadic λ for which each retained segment J meets X in volume comparable to $\lambda|T|$. Select a dyadic number β of such segments per retained tube. Necessarily $1 \le \beta \lesssim r^\alpha$. Let

$$Y(T) = \bigcup_{J \text{ selected for } T} (J \cap X \cap T)$$

with grid-cell shadings in a fixed tube enlargement. Its density is comparable to $\lambda_0 = \beta\lambda$. On a fixed common time interval the truncated sum is exactly the extension of the subfamily whose interval is active, modulo the already controlled spatial tails. There is no pointwise monotonicity assertion when deleting packets.

The selected broad measure is carried by a set $X_2 \subset X$, with

$$|X_2| \gtrsim (\log r)^{-C}|X|, \tag{4.17}$$

and with comparable selected broad mass on its constituent cubes. The broad parameter after these selections is at least $A/(\log r)^C$.

Case 1: few active time intervals. Suppose $\beta \le r^{b_0}$. Partition each common time interval spatially into cubes of side comparable to r_1. For such a cube B_j, let $\boldsymbol{f}_j$ be the vector of active packets whose tubes meet a fixed enlargement of B_j, retaining sector labels. On B_j the truncated sum agrees with $E\boldsymbol{f}_j$ up to rapidly decreasing errors. A tube meets only $O_n(1)$ spatial cubes in each active interval, since its slope is bounded and its width is $o(r_1)$. Consequently,

$$\begin{aligned} \sum_j \|\boldsymbol{f}_j\|_2^2 &\lesssim r^{b_0+C\varepsilon_0}\|\boldsymbol{f}\|_2^2, \\ \max_{\theta:\, r_1{}^{-1/2}\text{-cap}} \|\boldsymbol{f}_j\|_{L^2_{\text{avg}}(\theta)} &\lesssim \max_{\theta:\, r^{-1/2}\text{-cap}} \|\boldsymbol{f}\|_{L^2_{\text{avg}}(\theta)}. \end{aligned} \tag{4.18}$$

The average estimate follows from (2.17); the new support buffer absorbs the packet-projection enlargement.

Since

$$\frac{r^{a_0}}{(\log r)^C} \ge r^{(1-\alpha)a_0} = r_1^{a_0}$$

for sufficiently large R, the induction hypothesis applies on every B_j. After summation the ratio to the right side of (4.16) is bounded by

$$C(\log r)^C r^{-\eta\alpha+b_0+C\varepsilon_0}.$$

The parameter choices make this smaller than $1/2$. This closes this case, including all constants.

Case 2: many active time intervals. Suppose $\beta > r^{b_0}$. A ball of radius comparable to r_1 meets only $O_n(1)$ of the common time intervals. Since all selected segments have comparable shading size, its share of any $Y(T)$ is at most C/β. Hence the normalized shading is a two-ends shading with parameters comparable to 2α and b_0 at tube scale $\delta \asymp r^{-1/2}$, with the small enlargement adjustments already budgeted.

Apply (2.5) to this *ancestor* shading before selecting transverse caps. This gives a union of good grid cubes with exceptional volume at most $r^{-c\alpha}|X|$ and the multiplicity bound (4.6). By (4.17) and the uniform broad mass on X_2, the exceptional set removes less than half of the selected broad mass for sufficiently large R. Restrict to the good set. No further two-ends property is required of subsequent restrictions.

Apply Lemma 2.7. Fix a transverse tuple at cost K^C, then one spatial cube in one common time interval at cost at most $r^{C_n\alpha}$. On that cube the active packet families are fixed complete extensions. Pigeonhole their separate positive multiplicities on the $r^{1/2+\varepsilon_0}$ grid while retaining the additive functional $\sum_{b\subset X}\min_i\int_b|G_i|^{p_b}$ from Lemma 2.7; there are only polylogarithmically many joint bins. Choose a bin carrying a polylogarithmic fraction of that functional. On that bin, and not before it is chosen, bound the functional by $(\prod_i\int_X|G_i|^{p_b})^{1/k}$. Cubes where a family has zero multiplicity contribute only controlled tails to the local minimum functional and are removed. The surviving families meet the hypotheses of Subsection 4.3, with the fixed subfamily in the selected time slab used for each G_i. Their average densities are no greater than a constant times the ancestor density, since every packet's new shading is a subset of its ancestor shading. The inherited good-set upper bound is unchanged.

If a multiplicity is small, use the direct low-multiplicity argument there. Otherwise use Lemma 4.2 and (4.13). Combining all factors yields

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \lesssim_\eta K^{C_{n,k}} r^{C_{n,k}\alpha+o_{\varepsilon_0}(1)+\varepsilon'}(\log r)^C\|f\|_2^2\max_{\theta:\,r^{-1/2}\text{-cap}}\|f\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)},$$

where the arbitrary input loss $\varepsilon'>0$ was chosen sufficiently small. The parameter choices absorb this into $C_\eta R^\eta r^\eta\|f\|_2^2\max_{\theta:\,r^{-1/2}\text{-cap}}\|f\|^{p_b-2}_{L^2_{\mathrm{avg}}(\theta)}$. The constant C_η can be enlarged once to cover the base scales. This completes the induction and proves (4.15). □

Remark 4.5 (What the parameter bookkeeping fixes). The parameter A has not been declared fixed while being divided by arbitrarily many logarithms. The induction keeps K tied to the fixed outer radius R, stops at R^θ, and only decreases the intermediate radius r. The factor $r^{a_0\alpha}$ between successive allowed lower bounds for A pays for every polylogarithmic division. The cost of the resulting $A=R^{a_0}$ must still be included in the narrow argument; it is not hidden in a constant.

4.5. **From mixed broad estimates to linear restriction.** Fix

$$\max\{p_b, p_N\} < p < \frac{2(k-1)}{k-2}, \tag{4.19}$$

By Proposition 4.4, and the bound $\max_{\theta:\, R^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)} \le \|f\|_\infty$, we have

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} \lesssim_{\eta,p} R^{2\eta}\|f\|_2^2\|f\|_\infty^{p-2}, \qquad K \asymp R^\kappa, \quad A \ge R^{a_0}. \tag{4.20}$$

We use this estimate directly in the narrow induction, with target

$$\int_{B_R} |Ef|^p \le C_\varepsilon R^\varepsilon \|f\|_2^2\|f\|_\infty^{p-2}. \tag{4.21}$$

The broad–narrow decomposition and [Guth, Lemma 9.3] give, for every $\delta_N > 0$,

$$\int_{B_R} |Ef|^p \lesssim K^{C_{n,p}}\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} + C_{\delta_N} A^{p-1} K^{\delta_N + (k-2)(p/2-1)} \sum_\tau \int_{B_{2R}} |Ef_\tau|^p. \tag{4.22}$$

Here each narrow cap is assigned to one exceptional plane; Hölder over the resulting disjoint groups accounts for A^{p-1}.

For a cap τ of radius K^{-1}, let $g_\tau(\xi) = f_\tau(\xi_\tau + K^{-1}\xi)$. Then

$$\|g_\tau\|_2^2 = K^{n-1}\|f_\tau\|_2^2, \qquad \|g_\tau\|_\infty = \|f_\tau\|_\infty.$$

Parabolic rescaling maps B_{2R} into a ball of radius CR/K. Applying the induction hypothesis there yields

$$\int_{B_{2R}} |Ef_\tau|^p \lesssim C_\varepsilon (R/K)^\varepsilon K^{2n-(n-1)p}\|f_\tau\|_2^2\|f_\tau\|_\infty^{p-2}. \tag{4.23}$$

Since the frequency cutoffs have bounded overlap,

$$\sum_\tau \|f_\tau\|_2^2\|f_\tau\|_\infty^{p-2} \lesssim \|f\|_2^2\|f\|_\infty^{p-2}.$$

Consequently, setting

$$\nu_p = \frac{2n-k}{2}p - (2n-k+2) > 0, \tag{4.24}$$

the narrow contribution in (4.22) is bounded by

$$CC_\varepsilon R^\varepsilon A^{p-1} K^{-\nu_p + \delta_N - \varepsilon}\|f\|_2^2\|f\|_\infty^{p-2}. \tag{4.25}$$

Choose the parameters in Proposition 4.4 so that

$$2\eta + \kappa C_{n,p} < \frac{\varepsilon}{2}, \qquad \delta_N < \frac{\nu_p}{4}, \qquad \frac{a_0(p-1)}{\kappa} < \frac{\nu_p}{4},$$

and take $A = \lceil R^{a_0} \rceil$. This is possible since a_0 may be chosen arbitrarily small after κ is fixed. The factor $A^{p-1}K^{-\nu_p+\delta_N-\varepsilon}$ is then $O(K^{-\nu_p/2})$, so the narrow term is absorbed into (4.21) for sufficiently large R. Meanwhile, (4.20) bounds the broad term by

$$C_{\eta,p} R^{\varepsilon/2}\|f\|_2^2\|f\|_\infty^{p-2}.$$

Taking C_ε sufficiently large to cover this term and the bounded base scales closes the induction. Finally, restricted-type real interpolation and a standard ϵ-removal argument (see [Tao99]) complete the proof.

4.6. **Limits of the present framework.** The broad–narrow argument in our framework is, to some extent, a substitute for the planebrush method, whose higher-dimensional analogue has not yet been established. If a suitable higher-dimensional planebrush estimate becomes available, it could potentially be incorporated into the Wang–Wu framework, leading to stronger restriction estimates than those obtained by our multilinear approach, particularly in high dimensions. Thus, the present framework should be viewed as an intermediate approach rather than a definitive improvement over the Wang–Wu method.

References

[BCCLXZ] T. Borges, T. Chan, M. Chen, D. Liu, Y. Xi, and Y. Zhan, *Restriction and Kakeya maximal estimates in* $\mathbb{R}^4$, preprint, arXiv:2511.22824v1 (2025).

[BG] J. Bourgain and L. Guth, *Bounds on oscillatory integral operators based on multilinear estimates*, Geom. Funct. Anal. **21** (2011), 1239–1295.

[DW] C. Demeter and S. Wu, *Restriction and decoupling estimates for the hyperbolic paraboloid in* $\mathbb{R}^3$, preprint, arXiv:2505.09037v2 (2025).

[GWX] C. Gao, S. Wu, and Y. Xi, *Sharp microlocal Kakeya–Nikodym estimates for eigenfunctions with applications*, preprint, arXiv:2509.01116v4 (2026).

[GWZ] S. Guo, H. Wang, and R. Zhang, *A dichotomy for Hörmander-type oscillatory integral operators*, Invent. Math. **238** (2024), 503–584. The displayed benchmark uses Theorem 4.2 in arXiv:2210.05851v2.

[Guth] L. Guth, *Restriction estimates using polynomial partitioning II*, J. Amer. Math. Soc. **31** (2018), 81–115.

[GIOW] L. Guth, A. Iosevich, Y. Ou, and H. Wang, *On Falconer's distance set problem in the plane*, preprint version, arXiv:1808.09346 (2018).

[HRZ] J. Hickman, K. M. Rogers, and R. Zhang, *Improved bounds for the Kakeya maximal conjecture in higher dimensions*, preprint, arXiv:1908.05589 (2019).

[HZ] J. Hickman and J. Zahl, *A note on Fourier restriction and nested polynomial Wolff axioms*, J. Anal. Math. **152** (2024), 19–52.

[KT] N. H. Katz and T. Tao, *New bounds for Kakeya problems*, J. Anal. Math. **87** (2002), 231–263. The precise two-ends formulation used here is recorded in [WW, Section 3.2].

[KZ] N. H. Katz and J. Zahl, *A Kakeya maximal function estimate in four dimensions using planebrushes*, preprint version, arXiv:1902.00989.

[Moreira] T. Moreira, *Refinements on the Restriction and Kakeya problems range in* $\mathbb{R}^4$, preprint, arXiv:2610.06181v1 (2026).

[RS] C. Rios and E. T. Sawyer, *The Fourier extension conjecture for the paraboloid*, preprint, arXiv:2512.24990v2 (2026). This item is noted for the separate literature-priority audit, not used as an input.

[Tao99] T. Tao, *The Bochner–Riesz conjecture implies the restriction conjecture*, Duke Math. J. **96** (1999), 363–376.

[WW] H. Wang and S. Wu, *Restriction estimates using decoupling theorems and two-ends Furstenberg inequalities*, preprint, arXiv:2411.08871v3 (2024).

[Zahl] J. Zahl, *New Kakeya estimates using Gromov's algebraic lemma*, Adv. Math. **380** (2021), 107596.

Department of Mathematics, University of Illinois Urbana-Champaign
Email address: xw70@illinois.edu